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## COMPLEX NUMBERS

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In complex numbers, the operation rule for the imaginary unit  $i$  is:

$$i^2 = -1$$

If the real numbers are on an axis, then the complex numbers are in the complex plane. Any complex number  $z = x + iy$  ( $x$  and  $y$  real) can be represented by a point  $(x, y)$  in the  $(x, y)$  plane.

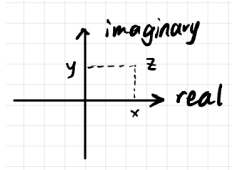


Figure 1: The complex plane

## 1. Real-imaginary part representation

### 1.1 $+, -, \times, \div$ of complex numbers

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 - z_2 = (x_1 + x_2) + i(y_1 - y_2)$$

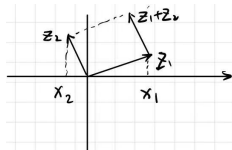


Figure 2: Addition of complex numbers

$$\begin{aligned} z_1 \cdot z_2 &= (x_1 + iy_1) \cdot (x_2 + iy_2) \\ &= x_1 \cdot x_2 + ix_1 \cdot y_2 - y_1 \cdot y_2 + ix_2 \cdot y_1 \\ &= (x_1 \cdot x_2 - y_1 \cdot y_2) + i(x_1 \cdot y_2 + ix_2 \cdot y_1) \end{aligned}$$

In order to understand complex division, we need to first understand complex conjugation.  $x + iy$  and  $x - iy$  are conjugate

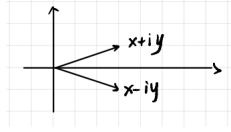


Figure 3: Conjugate complex numbers

Magnitude of a complex number  $z = x + iy$ :

$$|z| = \sqrt{x^2 + y^2}$$

Using complex conjugation to compute complex division:

$$(x + iy) \cdot (x - iy) = x^2 + y^2$$

Example:

$$\frac{2+i}{3-i} = \frac{(2+i) \cdot (3+i)}{(3-i) \cdot (3+i)} = \frac{6-1+5i}{10} = \frac{1}{2} + i\frac{1}{2}$$

### 1.2 Some operations on the conjugation of complex numbers

$$\overline{(Z_1 + Z_2)} = \overline{Z_1} + \overline{Z_2}$$

$$\overline{Z_1 \cdot Z_2} = \overline{Z_1} \cdot \overline{Z_2}$$

$$\frac{\overline{Z_1}}{\overline{Z_2}} = \frac{\overline{Z_1}}{\overline{Z_2}}$$

$$\text{if } P(z) = a_2 z^2 + a_1 z + a_0, \text{ then } \overline{P(z)} = \overline{a_2 z^2 + a_1 z + a_0}$$

## 2. Phase-magnitude representation

Complex number  $z = re^{i\theta}$

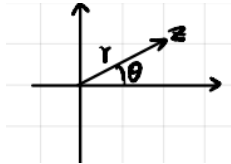


Figure 4: Phase-magnitude representation

## 2.1 Complex number operations under phase-magnitude representation

$$\begin{aligned}
 e^{i\theta} &= \cos \theta + i \sin \theta \\
 re^{i\theta} &= r \cos \theta + ir \sin \theta \\
 z_1 \cdot z_2 &= (r_1 \cdot r_2) \cdot e^{i(\theta_1 + \theta_2)} \\
 z_1 / z_2 &= \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \\
 \bar{z} &= re^{-i\theta} \\
 z \cdot \bar{z} &= r^2 e^{i\theta} \cdot e^{-i\theta} = r^2
 \end{aligned}$$

## 2.2 Complex equations

Example 1:

$$\begin{aligned}
 (x + iy)^2 &= 2i \\
 x^2 - y^2 + i2xy &= 2i \\
 \begin{cases} x^2 - y^2 = 0 & x = y = 1 \\ 2xy = 2 & x = y = 1 \end{cases}
 \end{aligned}$$

Example 2:

$$\begin{aligned}
 |x + iy| &= y - ix \\
 \sqrt{x^2 + y^2} &= y - ix \\
 x^2 + y^2 &= (y - ix)^2
 \end{aligned}$$

### Euler formula

$$\begin{aligned}
 e^{i\theta} &= \cos \theta + i \sin \theta = \cos \theta + i \sin \theta \\
 e^{-i\theta} &= \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta \\
 \Rightarrow \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2} \\
 \Rightarrow \sin \theta &= \frac{e^{i\theta} - e^{-i\theta}}{2i} \\
 \text{eg. } \int_{-\infty}^{\infty} x^{-2} \cos x dx &= \int_{-\infty}^{\infty} x^{-2} \cdot \frac{e^{ix} + e^{-ix}}{2} dx
 \end{aligned}$$

### Some function of a complex variable

$$f(z) = 2z^2 + z + 1, \frac{z^2 + 1}{z - 3}, z^n, z^{1/2}, \sqrt[3]{z}$$

1. polynomial function

$$f(z) = \frac{z^2 + 1}{z - 3}$$

Example 1:

if  $z = i - 2$ :

$$f(z)|_{z=i-2} = \frac{(i-2)^2 + 1}{(i-2)^2 - 3}$$

2. exponential function

$$e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y)$$

Example 2:

$$\begin{aligned} e^{-i} &= \cos(-1) + i \sin(-1) = \cos 1 - i \sin 1 \\ e^{2-i\pi} &= e^2(\cos \pi + i \sin \pi) = e^2[(-1) + 0] = -e^2 \end{aligned}$$

3. trigonometric function

$$\begin{aligned} \cos z &= \frac{e^{iz} + e^{-iz}}{2} \\ \sin z &= \frac{e^{iz} - e^{-iz}}{2i} \end{aligned}$$

Example 3:

$$\begin{aligned} \cos i &= \frac{e^{i \cdot i^2} + e^{-i \cdot i}}{2} = \frac{e^{-1} + e}{2} \\ \sin\left(\frac{\pi}{2} + i \ln 2\right) &= \frac{e^{i\left(\frac{\pi}{2} + i \ln 2\right)} - e^{-i\left(\frac{\pi}{2} + i \ln 2\right)}}{2i} = \frac{e^{-\ln 2 + i\frac{\pi}{2}} - e^{\ln 2 - i\frac{\pi}{2}}}{2i} \\ \cos^2 x + \sin^2 x &= 1 \Rightarrow \cos^2 z + \sin^2 z = 1 \end{aligned}$$

4. Hyperbolic functions

$$\begin{aligned} z = iy \quad \cos iy &= \frac{e^{i \cdot iy} + e^{-i \cdot iy}}{2} = \frac{e^{-y} + e^y}{2} \\ \Rightarrow \cosh z &= \frac{e^z + e^{-z}}{2} \quad \sinh z = \frac{e^z - e^{-z}}{2} \end{aligned}$$